Calcium gluconate: Drug information - UpToDate

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Contributor Disclosures

For additional information see "Calcium gluconate: Patient drug information" and "Calcium gluconate: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: Canada

- Calcium Sandoz

Pharmacologic Category

- Calcium Salt;
- Electrolyte Supplement, Oral;
- Electrolyte Supplement, Parenteral

Dosing: Adult

Dosage guidance:

Dosing: Dosages are expressed in terms of the calcium gluconate salt (unless otherwise specified as elemental calcium). Dosages expressed in terms of the calcium gluconate salt are based on a solution concentration of 100 mg/mL (10%) containing 0.465 mEq (9.3 mg) elemental calcium per mL, except where noted. One gram of calcium gluconate salt is equal to 93 mg of elemental calcium.

Beta-blocker toxicity

Beta-blocker toxicity (off-label use): Based on limited data: **IV:** Initial: 3 to 6 **g** (30 to 60 **mL** of a 10% solution) or 60 **mg/kg** (0.6 **mL/kg** of a 10% solution) over 5 to 10 minutes; may repeat every 10 to 20 minutes for 3 to 4 additional doses **or** initiate a continuous infusion of 60 to 120 **mg/kg/hour** titrated to improve hemodynamic response (Ref).

Calcium channel blocker toxicity

Calcium channel blocker toxicity (off-label use): Based on limited data: **IV:** Initial: 3 to 6 **g** (30 to 60 **mL** of a 10% solution) or 60 **mg/kg** (0.6 **mL/kg** of a 10% solution) over 5 to 10 minutes; may repeat every 10 to 20 minutes for 3 to 4 additional doses **or** initiate a continuous infusion of 60 to 120 **mg/kg/hour** titrated to improve hemodynamic response (Ref). In one report, 18 **g** was administered over a 3-hour period (Ref). **Note:** Maintain serum ionized calcium at 1.5 to 2 times the ULN (Ref).

Calcium replacement, during CRRT with citrate-based regional anticoagulation

Calcium replacement, during CRRT with citrate-based regional anticoagulation (off-label use) :

Intracircuit, posthemodialysis filter (returning to the patient) or IV through a separate central line: Ensure adequate systemic ionized calcium concentrations (drawn directly from the patient) prior to initiation of CRRT. During CRRT, use a calcium gluconate continuous infusion sliding scale to maintain systemic ionized calcium between ~3.6 to 5.2 mg/dL (~0.9 to 1.3 mmol/L); monitor systemic ionized calcium every 6 hours (or more frequently when indicated) (Ref). Refer to institutional protocols.

Hydrofluoric acid exposure

Hydrofluoric acid exposure (off-label use):

Systemic toxicity, prevention and treatment: **Note:** Profound and precipitous hypocalcemia may occur after exposure to higher hydrofluoric acid concentrations to even a small surface area (Ref). Additional treatment measures may be required (eg, magnesium) (Ref); consultation with a clinical toxicologist or poison center is highly recommended.

IV: 2 g immediately or as soon as possible after exposure (before serum calcium level is known). Monitor serum calcium level; if serum calcium level is not within normal range, then give an additional 1 g, followed by 4 g over 1 hour, as needed, to maintain serum calcium levels (Ref).

Hydrofluoric acid burns, treatment:

Topical: 2.5% gel: After washing the affected area, apply liberally and allow the application to stay on the affected area for ≥30 minutes or for ≥15 minutes past the point of pain resolution, whichever is longer; cover and redress every 4 hours as needed. **Note:** Surgical gloves filled with gel are useful for hand burns (Ref). The use of gel-filled condom for penile exposure has been reported (Ref).

Note: If commercially available or extemporaneously compounded gel is unavailable, then topical application of wet compresses made with the 10% calcium gluconate IV solution may be used immediately following decontamination (Ref).

SUBQ (off-label route): 5% to 10% solution: 0.5 **mL**/cm^2 of burned tissue (Ref). Infiltration should be carried 0.5 cm away from the margin of the injured tissue into the surrounding uninjured areas. Repeat if pain recurs. Local anesthesia may be required to perform procedure; pain resolution is the therapeutic endpoint and if a local anesthetic is utilized, it may be difficult to determine the success of therapy (**Note: Never** use calcium chloride for SUBQ injection as it may result in severe dermal necrosis).

Intra-arterial (off-label route): Add 1 g (10 **mL** of a 10% solution) to 50 mL of D5W. Infuse over 4 hours into the artery that provides the vascular supply to the affected area (Ref). Pain usually resolves by the end of the infusion; repeat if pain recurs. **This intervention should be used only by those accustomed to this technique. Extreme care should be taken to avoid the extravasation.** A poison information center or clinical toxicologist should be consulted prior to implementation.

IV (Bier block technique) (off-label route): Add 1.5 g (15 **mL** of a 10% solution) to 35 mL of NS and infuse over 2 minutes using a Bier block technique (Ref).

Hydrofluoric acid inhalational exposure, treatment: **Inhalation (off-label route):** 2.5% nebulization solution: Mix 1.5 **mL** of 10% calcium gluconate solution with 4.5 mL NS to make a 2.5% solution and administer via nebulization; repeat doses may be given after medical evaluation (Ref).

Hyperkalemia, severe/emergent

Hyperkalemia , severe/emergent (off-label use): Note: Use in patients with hyperkalemia-associated ECG changes **or** serum potassium >6.5 mEq/L (Ref). Stabilizes myocardial cell membrane without impacting plasma potassium concentrations; must combine with insulin/dextrose to decrease plasma potassium levels and other therapies to eliminate potassium from body (Ref). Perform continuous cardiac monitoring and obtain serial ECGs (Ref).

IV, Intraosseous: Initial: 1 to 2 g over 2 to 5 minutes; may repeat after 5 minutes if ECG changes persist or recur, then every 30 to 60 minutes as needed; usual initial dosage range: 1 to 3 g (Ref).

Hypocalcemia

Hypocalcemia:

Note: For use in patients with severe symptoms of hypocalcemia (eg, tetany, seizures, carpopedal spasm), ECG abnormalities (eg, QTc prolongation, arrhythmia), or an acute decrease in albumin-corrected serum calcium levels to <7 to 7.5 mg/dL (<1.75 to 1.87 mmol/L) when serious complications may occur if untreated (eg, following neck surgery). Should also be used in patients whose symptoms did not improve with or who can no longer take oral calcium. Correct concurrent hypomagnesemia if present (Ref). **Do not** use IV calcium as initial therapy in patients with chronic kidney disease who are asymptomatic or who have stable hypocalcemia with only mild symptoms (eg, paresthesias) (Ref).

Initial bolus dose(s): **IV:** Add 1 or 2 g (10 or 20 **mL** of a 10% solution) to 50 mL of D5W or NS (equivalent to ~90 or ~180 mg **elemental calcium**). Infuse over 10 to 20 minutes; may repeat bolus dose after 10 to 60 minutes if symptoms persist (Ref). If hypocalcemia is expected to persist (eg, hypoparathyroidism, pancreatitis), follow bolus dose(s) with a continuous IV calcium infusion (Ref). **Note:** In patients with chronic hypocalcemia who require IV therapy, may initiate therapy with a continuous IV infusion and omit bolus dosing (Ref).

Continuous infusion: **IV :** Add 11 g (110 **mL** of a 10% solution) to 890 mL of D5W or NS (equivalent to ~1 g **elemental calcium** in a final total volume of 1,000 mL). Initiate infusion at 50 to 100 mL/hour (equivalent to ~50 to 100 mg/hour of **elemental calcium**; adjust dose to maintain albumin-corrected serum calcium levels at the low end of normal. Initiate oral calcium and vitamin D supplements as soon as possible; once an effective oral regimen is achieved, taper IV calcium infusion slowly (eg, over 24 to 48 hours) (Ref).

Parenteral nutrition, maintenance requirement

Parenteral nutrition, maintenance requirement: Note: Expressed in terms of **elemental calcium: IV:** 10 to 15 **mEq elemental calcium** daily (Ref).

Radium-226 or strontium-90 internal contamination

Radium-226 or strontium-90 internal contamination (off-label use): IV: 2.5 g in 500 mL D5W over 4 to 6 hours once daily for up to 6 days; if possible, administer within 12 hours of radionuclide intake (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

Note: Do not use IV calcium as initial therapy in patients with chronic kidney disease who are asymptomatic or who have stable hypocalcemia with only mild symptoms (eg, paresthesias) (Ref).

Initiate with the lower limit of the dosage range (accumulation may occur with kidney impairment and subsequent doses may require adjustment based on serum calcium concentrations).

Dosing: Liver Impairment: Adult

No initial dosage adjustment necessary; subsequent doses should be guided by serum calcium concentrations. In patients in the anhepatic stage of liver transplantation, equal rapid increases in ionized concentrations occur suggesting that calcium gluconate does not require hepatic metabolism for release of ionized calcium (Ref).

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see ”Calcium gluconate: Pediatric drug information”)

Dosage guidance:

Dosing: Dosages are expressed in terms of the calcium gluconate salt or elemental calcium; use caution.

1 gram calcium gluconate = 93 mg elemental calcium = 4.65 mEq calcium.

Calcium channel blocker toxicity, hypotension/conduction disturbances

Calcium channel blocker toxicity, hypotension/conduction disturbances: Infants, Children, and Adolescents: Dose expressed as **calcium gluconate**: IV: 60 mg/kg/**dose** administered over 30 to 60 minutes; maximum dose: 3,000 to 6,000 mg/**dose** (Ref); some experts recommend rapid administration (eg, infuse over 5 to 10 minutes) and recommend to repeat doses every 10 to 20 minutes up to 3 to 4 doses as needed (Ref); may also consider IV infusion (Ref). **Note:** Calcium chloride may provide a more rapid increase of ionized calcium in critically ill children. Calcium gluconate may be substituted if calcium chloride is not available or if administering through a peripheral line (Ref).

Hydrofluoric acid burns, treatment

Hydrofluoric acid burns, treatment: Limited data available: Children and Adolescents:

SUBQ: 5% to 10% solution: 0.5 **mL**/cm^2 of burned tissue (Ref). Infiltration should be carried 0.5 cm away from the margin of the injured tissue into the surrounding uninjured areas. Repeat if pain recurs. Local anesthesia may be required to perform procedure; pain resolution is the therapeutic endpoint and if a local anesthetic is utilized, it may be difficult to determine the success of therapy. **Note: Never** use calcium chloride for subcutaneous injection.

Intra-arterial: Add 10 **mL** of a 10% solution to 50 mL of D5W. Infuse over 4 hours into the artery that provides the vascular supply to the affected area (Ref). Pain usually resolves by the end of the infusion; repeat if pain recurs. **This intervention should be used only by those accustomed to this technique. Extreme care should be taken to avoid the extravasation.** A poison information center or clinical toxicologist should be consulted prior to implementation.

Inhalation: 2.5% nebulization solution: Mix 1.5 **mL** of 10% calcium gluconate solution with 4.5 mL NS to make a 2.5% solution and administer via nebulization (Ref).

Hyperkalemia, adjunctive treatment

Hyperkalemia, adjunctive treatment:

Note: Calcium is administered for myocardium stabilization and prevention of arrhythmias; must be used in conjunction with therapy to shift potassium intracellularly and/or eliminate potassium.

Infants, Children, and Adolescents: Dose expressed as **calcium gluconate**: IV, Intraosseous: 60 to 100 mg/kg/**dose**; maximum dose: 2,000 mg/**dose**; may repeat if necessary (Ref).

Hypocalcemia, treatment

Hypocalcemia, treatment: Dose depends on clinical condition and serum calcium concentration; monitor closely.

Asymptomatic: Limited data available: Infants, Children, and Adolescents: Dose expressed as **elemental calcium**: Oral: 30 to 75 mg/kg/day in 4 to 5 divided doses (Ref). **Note:** In general, other oral calcium salts (eg, carbonate, glubionate) are a more preferable oral dosage form option in young pediatric patients; however, the 10% calcium gluconate injection may be given orally (Ref).

Mild to moderate symptoms:

IV Intermittent:

Infants, Children, and Adolescents <17 years: Dose expressed as **calcium gluconate**: IV: 29 to 60 mg/kg/**dose** every 6 hours (Ref).

Adolescents ≥17 years: Dose expressed as **calcium gluconate**: IV: 1,000 to 2,000 mg/**dose** every 6 hours (Ref).

Continuous IV Infusion:

Infants, Children, and Adolescents <17 years: Initial: Dose expressed as **calcium gluconate**: IV: 8 to 13 mg/kg/hour (Ref).

Adolescents ≥17 years: Initial: Dose expressed as **calcium gluconate**: IV: 5.4 to 21.5 mg/kg/hour (Ref).

Severe symptoms (eg, seizures, tetany): Infants, Children, and Adolescents: Dose expressed as **calcium gluconate**: IV, Intraosseous: 100 to 200 mg/kg/**dose** over 5 to 10 minutes, maximum dose: 1,000 to 2,000 mg/**dose**; may repeat as needed or follow with a continuous IV infusion of 8 to 32 mg/kg/hour (Ref).

Parenteral nutrition, maintenance calcium requirement

Parenteral nutrition, maintenance calcium requirement (Ref): **Note:** Dose expressed as **elemental calcium:**

Infants and Children ≤50 kg: IV: 0.5 to 4 **mEq**/kg/day as an additive to parenteral nutrition solution.

Children >50 kg and Adolescents: IV: 10 to 20 **mEq**/day as an additive to parenteral nutrition solution.

Rickets, treatment

Rickets, treatment: Limited data available: **Note:** Treatment should also include adequate vitamin D supplementation (Ref).

Infants, Children, and Adolescents: Dose expressed as **elemental calcium**: Oral: 30 to 75 mg/kg/day in 3 divided doses; begin at higher end of range and titrate downward over 2 to 4 weeks (Ref). **Note:** In general, other oral calcium salts (eg, carbonate, glubionate) are a more preferable oral dosage formulation option in young pediatric patients; however, the 10% calcium gluconate injection may be given orally (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

Initiate at the lowest dose of the recommended dosage range; monitor serum calcium concentrations closely. Accumulation may occur with renal impairment and subsequent doses may require adjustment based on serum calcium concentrations.

Dosing: Liver Impairment: Pediatric

No initial dosage adjustment necessary; subsequent doses should be guided by serum calcium concentrations. Hepatic function does not affect the availability of ionized calcium after calcium gluconate administration.

Adverse Reactions

The following adverse drug reactions are derived from product labeling unless otherwise specified.

Postmarketing:

Cardiovascular: Bradycardia, cardiac arrhythmia, decreased blood pressure, syncope, vasodilation

Dermatologic: Cutaneous calcification

Local: Inflammation at injection site, tissue necrosis at injection site

Contraindications

Hypercalcemia; concomitant use of IV calcium gluconate with ceftriaxone in neonates (≤28 days of age).

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Warnings/Precautions

Concerns related to adverse effects:

• Extravasation: Parenteral calcium is a vesicant; ensure proper catheter or needle position prior to and during infusion. Avoid extravasation; adverse events from extravasation can be devastating (eg, profound tissue necrosis). Monitor the IV site closely.

• GI effects: Constipation, bloating, and gas are common with oral calcium supplements (especially carbonate salt).

Disease-related concerns:

• Hyperphosphatemia: Use with caution in patients with severe hyperphosphatemia as elevated levels of phosphorus and calcium may result in soft tissue and pulmonary arterial calcium-phosphate precipitation.

• Hypokalemia: Use with caution in patients with severe hypokalemia as acute rises in serum calcium levels may result in life-threatening cardiac arrhythmias.

• Hypomagnesemia: Hypomagnesemia is a common cause of hypocalcemia; therefore, correction of hypocalcemia may be difficult in patients with concomitant hypomagnesemia. Evaluate serum magnesium and correct hypomagnesemia (if necessary), particularly if initial treatment of hypocalcemia is refractory.

• Kidney stones (calcium-containing): Use caution when administering calcium supplements to patients with a history of kidney stones.

• Renal impairment: Use with caution in patients with chronic renal failure to avoid hypercalcemia; frequent monitoring of serum calcium and phosphorus is necessary.

Dosage form specific issues:

• Aluminum: The parenteral product may contain aluminum; toxic aluminum concentrations may be seen with high doses, prolonged use, or renal dysfunction. Premature neonates are at higher risk due to immature renal function and aluminum intake from other parenteral sources. Parenteral aluminum exposure of >4 to 5 mcg/kg/day is associated with CNS and bone toxicity; tissue loading may occur at lower doses (Federal Register 2002). See manufacturer’s labeling.

• Appropriate product selection: Multiple salt forms of calcium exist; close attention must be paid to the salt form when ordering and administering calcium; incorrect selection or substitution of one salt for another without proper dosage adjustment may result in serious over or under dosing.

• IV administration: Avoid too-rapid IV administration (do not exceed 200 mg/minute in adults and 100 mg/minute in pediatric patients); may result in vasodilation, hypotension, bradycardia, arrhythmias, syncope, and cardiac arrest.

• Oral administration: Administering oral calcium with food and vitamin D will optimize calcium absorption.

• Topical administration: Avoid contact with eyes; not for oral administration.

• Tartrazine: Some products may contain tartrazine, which may cause allergic reactions in susceptible individuals.

Dosage Forms Considerations

1 g calcium gluconate = elemental calcium 93 mg = calcium 4.65 mEq = calcium 2.33 mmol

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous:

Generic: 10% (10 mL, 100 mL)

Solution, Intravenous [preservative free]:

Generic: 10% (10 mL, 50 mL, 100 mL); 1 g in NaCl 0.8% (100 mL); Calcium gluconate 1 g/50 mL in NaCl 0.675% (50 mL); Calcium gluconate 2 g/100 mL in NaCl 0.675% (100 mL)

Tablet, Oral:

Generic: 50 mg

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Calcium Gluconate Intravenous)

10% (per mL): $0.51 - $0.70

Solution (Calcium Gluconate-NaCl Intravenous)

1GM/100ML 0.8% (per mL): $0.15

1GM/50ML 0.675% (per mL): $0.23 - $0.47

2GM/100ML 0.675% (per mL): $0.24 - $0.47

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous:

Generic: 10% (10 mL, 50 mL, 100 mL, 500 mL); Calcium gluconate 1 g/50 mL in NaCl 0.675% (50 mL)

Extemporaneous Preparations

A calcium gluconate gel may be made with tablets or solution for injection. Crush seven calcium gluconate 500 mg tablets in a mortar and reduce to a fine powder and add a 5 oz (150 mL) tube of water-soluble surgical lubricant (eg, K-Y Jelly). Alternatively, add 2.5 g (25 mL) of calcium gluconate 10% solution for injection to 2.5 oz (75 mL) of water-soluble surgical lubricant (eg, K-Y Jelly). Calcium carbonate may be substituted; do not use calcium chloride due to the potential for irritation and further tissue injury (Ref).

Administration: Adult

Oral: Administer with plenty of fluids with or following meals.

IV: For bolus or continuous infusion. Administer bolus slowly (not to exceed 200 mg/minute) except in emergency situations; refer to indication-specific infusion rates in dosing for detailed recommendations. For continuous infusions, adjust rate as needed based on serum calcium levels. Typical rates of administration may vary with indication; refer to institutional protocol. Not for IM administration.

Vesicant; ensure proper needle or catheter placement prior to and during IV infusion. Avoid extravasation.

Extravasation management:

Early/acute calcium extravasation: If extravasation occurs, stop infusion immediately; leave needle/cannula in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution, then remove needle/cannula; elevate extremity; apply dry warm compresses; initiate hyaluronidase antidote (Ref).

Hyaluronidase: **Intradermal or SUBQ:** Inject a total of 1 to 1.7 mL (15 units/mL) as 5 separate 0.2 to 0.3 mL injections (using a tuberculin syringe) around the site of extravasation; if IV catheter remains in place, administer intravenously through the infiltrated catheter; may repeat in 30 to 60 minutes if no resolution (Ref).

Delayed calcium extravasation: Closely monitor site; most calcifications spontaneously resolve. However, if a severe manifestation of calcinosis cutis occurs, may initiate sodium thiosulfate antidote.

Sodium thiosulfate: IV: 12.5 g over 30 minutes, administered 3 times per week; may increase gradually to 25 g 3 times per week; monitor for non-anion gap acidosis, hypocalcemia, severe nausea (Ref).

Hydrofluoric acid exposure (off label):

Hydrofluoric acid burns, treatment:

SUBQ infiltration (off-label route): Using a 27- or 30-gauge needle, approach the wound from the distal point of injury and infiltrate directly into the affected dermis and subcutaneous tissue. The infiltration should be carried 0.5 cm away from the margin of the injured tissue into the surrounding uninjured areas (Ref). **Avoid excessive administration, as it can cause compartment syndrome and further exacerbate tissue damage.** Following subungual exposure, administer to the affected area via the lateral or volar route through the fat pad (under digital nerve block); administration may also require removal of the nailbed, splitting the distal nail from the nailbed, or trimming the nail to the nailbed to reach the affected area (Ref).

Topical: Massage into the affected area and cover with a dressing. For hand involvement, place hand in rubber glove filled with gel and bend fingers intermittently to ensure proper contact with gel. The use of gel-filled condom for penile exposure has been reported (Ref).

Intra-arterial (off-label route): Requires radiology to place an arterial catheter in an artery supplying blood to the area of exposure; infuse over four hours (Ref). **This intervention should be used only by those accustomed to this technique. Care should be taken to avoid the extravasation.** A poison information center or clinical toxicologist should be consulted prior to implementation.

Hydrofluoric acid inhalational exposure, treatment: *Inhalation (off-label route):* Mix 1.5 **mL** of 10% calcium gluconate solution with 4.5 mL NS to make a 2.5% solution and administer via nebulization (Ref).

Preparation for Administration: Adult

Note: Premixed solutions are available.

IV: Observe the vial or PAB container for the presence of particulates. If particulates are observed, place vial or PAB container in a 60°C to 80°C water bath with occasional agitation until solution is clear; shake vigorously; cool to room temperature before use. Do not use if particulates do not dissolve. Prior to administration, dilute in D5W or NS and use immediately:

Bolus: Dilute to a concentration of 10 to 50 mg/mL; may be given undiluted via central line during a cardiac arrest.

Continuous infusion: Dilute to a concentration of 5.8 to 10 mg/mL.

Note: Usual concentrations: 1 g/100 mL D5W or NS; 2 g/100 mL D5W or NS. Maximum concentration in parenteral nutrition solutions is variable depending upon concentration and solubility (consult detailed reference).

Inhalation: Treatment of hydrofluoric acid burns (off-label use): Mix 1.5 mL of 10% calcium gluconate solution with 4.5 mL NS to make a 2.5% solution (Ref).

Premixed solution: Do not further dilute; do not add additional medications to premixed solution. Observe bag for presence of particulates; discard if discolored or particulate matter is present.

Administration: Pediatric

Oral: Administer with plenty of fluids with or following meals. The 10% calcium gluconate injection may be administered orally in young pediatric patients (Ref).

Parenteral: Do not inject calcium salts IM or administer SUBQ since severe necrosis and sloughing may occur; extravasation of calcium can result in severe necrosis and tissue sloughing. Do not use scalp vein or small hand or foot veins for IV administration. Not for endotracheal administration.

Intermittent IV infusion: Vial must be diluted prior to administration; if using premix solution, do not dilute. Infuse dose slowly (not to exceed 100 mg/minute), except in emergency situations; administer through a secure IV line to avoid extravasation, dermal calcium deposits, or tissue necrosis. In acute situations of symptomatic hypocalcemia, infusions over 5 to 10 minutes have been described in pediatric patients (Ref).

Continuous IV infusion: Vial must be diluted prior to administration; if using premix solution, do not dilute. Adjust rate as needed based on serum calcium concentrations. Typical rates of administration may vary with indication; refer to institutional protocol.

Parenteral nutrition solution: Calcium-phosphate stability in parenteral nutrition solutions is dependent upon the pH of the solution, temperature, and relative concentration of each ion. The pH of the solution is primarily dependent upon the amino acid concentration. The higher the percentage amino acids the lower the pH, the more soluble the calcium and phosphate. Individual commercially available amino acid solutions vary significantly with respect to pH lowering potential and consequent calcium phosphate compatibility; consult product-specific labeling for additional information.

Vesicant; ensure proper needle or catheter placement prior to and during IV infusion. Avoid extravasation.

Early/acute calcium extravasation: If acute extravasation occurs, stop infusion immediately; leave needle/cannula in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution, then remove needle/cannula unless needed for IV antidote; elevate extremity; apply dry warm compresses; initiate hyaluronidase antidote (Ref).

Delayed calcium extravasation: If delayed extravasation suspected, closely monitor site; most calcifications spontaneously resolve. However, if a severe manifestation of calcinosis cutis occurs, may initiate sodium thiosulfate antidote (Ref).

Preparation for Administration: Pediatric

Parenteral:

Vial: Observe the vial for the presence of particulates. If particulates are observed, place vial in a 60°C to 80°C water bath for 15 to 30 minutes (or until solution is clear); occasionally shake to dissolve; cool to body/room temperature before use. Do not use vial if particulates do not dissolve.

IV infusion: Prior to administration, further dilute in D5W or NS and use immediately. Concentration is dependent on type of infusion.

Intermittent IV infusion: Dilute to 10 to 50 mg/mL.

Continuous IV infusion: Dilute to 5.8 to 10 mg/mL.

Parenteral nutrition: Maximum concentration in parenteral nutrition solutions is variable depending upon concentration and solubility (consult detailed reference).

Premixed solution: Do not further dilute. Observe bag for presence of particulates; discard if discolored or particulate matter is present.

Vesicant/Extravasation Risk

Vesicant

Storage/Stability

IV:

Premixed solution: Store at 20°C to 25°C (68°F to 77°F). Do not freeze. Use within 60 days of removal from overwrap. Discard any unused portion.

Vial, PAB container: Store at 20°C to 25°C (68°F to 77°F). Do not freeze. Protect PAB container from light until use. Discard unused portion within 4 hours after initial puncture.

Pharmacy supply of emergency antidotes: Guidelines suggest that at least 30 g of calcium gluconate be stocked. Suggested amount is stated to be a sufficient quantity to provide antidotal treatment for 1 patient weighing 100 kg for an initial 8- to 24-hour period (Dart 2018); actual amount to be stocked should take into account site-specific and population-specific needs.

Oral: Store at room temperature; consult product labeling for specific requirements.

Topical: Gel: Calgonate: 2.5% (25 g): Store at 15°C to 30°C (59°F to 85°F). Open tube only when needed; discard open or damaged tubes.

Use: Labeled Indications

Dietary supplement (oral only): Dietary calcium supplementation.

Hydrofluoric acid burns (Calgonate [topical]): Treatment of dermal hydrofluoric acid burns.

Hypocalcemia (IV only): Treatment of acute symptomatic hypocalcemia in adult and pediatric patients.

Use: Off-Label: Adult

Beta-blocker toxicity; Calcium channel blocker toxicity; Calcium replacement, during CRRT with citrate-based regional anticoagulation; Hyperkalemia, severe/emergent; Hydrofluoric acid exposure; Radium-226 or strontium-90 internal contamination

Medication Safety Issues

Sound-alike/look-alike issues:

Calcium gluconate may be confused with calcium glubionate, cupric sulfate

Administration issues:

Calcium gluconate may be confused with calcium chloride.

Confusion with the different intravenous salt forms of calcium has occurred. There is a threefold difference in the primary cation concentration between **calcium gluconate** (in which 1 g = 4.65 mEq [93 mg] of elemental Ca++) and **calcium chloride** (in which 1 g = 14 mEq [270 mg] of elemental Ca++).

Prescribers should specify which salt form is desired. To prevent medication errors, dosages should be expressed either as mg or grams of the salt form for most indications. Dosages expressed as mEq may be used for nutrition.

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Alpha-Lipoic Acid: Calcium Salts may decrease absorption of Alpha-Lipoic Acid. Alpha-Lipoic Acid may decrease absorption of Calcium Salts. Management: Separate administration of alpha-lipoic acid from that of any calcium-containing compounds by several hours. If alpha-lipoic acid is given 30 minutes before breakfast, then administer oral calcium-containing products at lunch or dinner. *Risk D: Consider Therapy Modification*

Baloxavir Marboxil: Polyvalent Cation Containing Products may decrease serum concentrations of Baloxavir Marboxil. *Risk X: Avoid*

Bictegravir: Calcium Salts may decrease serum concentrations of Bictegravir. Management: Bictegravir can be given with calcium under fed conditions. When fasting, coadministration with, or 2 hours after, calcium is not recommended. In pregnancy, bictegravir can be given 2 hours before or 6 hours after calcium under fasting conditions. *Risk D: Consider Therapy Modification*

Bisphosphonate Derivatives: Polyvalent Cation Containing Products may decrease serum concentrations of Bisphosphonate Derivatives. Management: Avoid administration of oral medications containing polyvalent cations within: 2 hours before or after tiludronate/clodronate/etidronate; 60 minutes after oral ibandronate; or 30 minutes after alendronate/risedronate. *Risk D: Consider Therapy Modification*

Cabotegravir: Polyvalent Cation Containing Products may decrease serum concentrations of Cabotegravir. Management: Administer polyvalent cation containing products at least 2 hours before or 4 hours after oral cabotegravir. *Risk D: Consider Therapy Modification*

Calcium Acetate: Calcium Salts may increase adverse/toxic effects of Calcium Acetate. *Risk X: Avoid*

Calcium Channel Blockers: Calcium Salts may decrease therapeutic effects of Calcium Channel Blockers. *Risk C: Monitor*

Calcium Polystyrene Sulfonate: Polyvalent Cation Containing Products may decrease therapeutic effects of Calcium Polystyrene Sulfonate. *Risk C: Monitor*

Cardiac Glycosides: Calcium Salts may increase arrhythmogenic effects of Cardiac Glycosides. *Risk C: Monitor*

CefTRIAXone: Calcium Salts (Intravenous) may increase adverse/toxic effects of CefTRIAXone. Ceftriaxone binds to calcium forming an insoluble precipitate. Management: Use of ceftriaxone is contraindicated in neonates (28 days of age or younger) who require (or are expected to require) treatment with IV calcium-containing solutions. In older patients, flush lines with compatible fluid between administration. *Risk X: Avoid*

Deferiprone: Polyvalent Cation Containing Products may decrease serum concentrations of Deferiprone. Management: Separate administration of deferiprone and oral medications or supplements that contain polyvalent cations by at least 4 hours. *Risk D: Consider Therapy Modification*

DOBUTamine: Calcium Salts may decrease therapeutic effects of DOBUTamine. *Risk C: Monitor*

Dolutegravir: Calcium Salts may decrease serum concentrations of Dolutegravir. Management: Administer dolutegravir at least 2 hours before or 6 hours after oral calcium. Administer dolutegravir/rilpivirine at least 4 hours before or 6 hours after oral calcium salts. Alternatively, dolutegravir and oral calcium can be taken together with food. *Risk D: Consider Therapy Modification*

Eltrombopag: Polyvalent Cation Containing Products may decrease serum concentrations of Eltrombopag. Management: Administer eltrombopag at least 2 hours before or 4 hours after oral administration of any polyvalent cation containing product. *Risk D: Consider Therapy Modification*

Elvitegravir: Polyvalent Cation Containing Products may decrease serum concentrations of Elvitegravir. Management: Administer elvitegravir-containing products 2 hours before, or 2 to 6 hours after, the administration of polyvalent cation containing products. *Risk D: Consider Therapy Modification*

Estramustine: Calcium Salts may decrease absorption of Estramustine. Management: Administer estramustine on an empty stomach, at least 1 hour before or 2 hours after the dose of an oral calcium supplement. If coadministered with calcium salts, monitor for decreased estramustine therapeutic effects. *Risk D: Consider Therapy Modification*

Levonadifloxacin: Calcium Salts may decrease serum concentrations of Levonadifloxacin. *Risk X: Avoid*

Multivitamins/Fluoride (with ADE): May increase serum concentrations of Calcium Salts. Calcium Salts may decrease serum concentrations of Multivitamins/Fluoride (with ADE). More specifically, calcium salts may impair the absorption of fluoride. Management: Avoid eating or drinking dairy products or consuming vitamins or supplements with calcium salts one hour before or after of the administration of fluoride. *Risk D: Consider Therapy Modification*

Multivitamins/Minerals (with ADEK, Folate, Iron): May increase serum concentrations of Calcium Salts. *Risk C: Monitor*

PenicillAMINE: Polyvalent Cation Containing Products may decrease serum concentrations of PenicillAMINE. Management: Separate the administration of penicillamine and oral polyvalent cation containing products by at least 1 hour. *Risk D: Consider Therapy Modification*

Phosphate Supplements: Calcium Salts may decrease absorption of Phosphate Supplements. Management: This applies only to oral phosphate and calcium administration. Administering oral phosphate supplements as far apart from the administration of an oral calcium salt as possible may be able to minimize the significance of the interaction. *Risk D: Consider Therapy Modification*

Quinolones: Calcium Salts may decrease absorption of Quinolones. Of concern only with oral administration of both agents. Management: Consider administering an oral quinolone at least 2 hours before or 6 hours after the dose of oral calcium to minimize this interaction. Monitor for decreased therapeutic effects of quinolones during coadministration. *Risk D: Consider Therapy Modification*

Raltegravir: Polyvalent Cation Containing Products may decrease serum concentrations of Raltegravir. Management: Administer raltegravir 2 hours before or 6 hours after administration of the polyvalent cations. Dose separation may not adequately minimize the significance of this interaction. *Risk D: Consider Therapy Modification*

Roxadustat: Polyvalent Cation Containing Products may decrease serum concentrations of Roxadustat. Management: Administer roxadustat at least 1 hour after the administration of oral polyvalent cation containing products. *Risk D: Consider Therapy Modification*

Strontium Ranelate: Calcium Salts may decrease serum concentrations of Strontium Ranelate. Management: Separate administration of strontium ranelate and oral calcium salts by at least 2 hours in order to minimize this interaction. *Risk D: Consider Therapy Modification*

Technetium Tc 99m Medronate: Coadministration of Calcium Salts and Technetium Tc 99m Medronate may alter diagnostic results. Management: Consider performing imaging with technetium Tc 99m medronate after cation levels have normalized (eg, 3 to 5 half-lives of the cation) in patients with high calcium levels caused by coadministration of calcium salts. *Risk D: Consider Therapy Modification*

Technetium Tc 99m Oxidronate: Coadministration of Calcium Salts and Technetium Tc 99m Oxidronate may alter diagnostic results. Management: Consider performing imaging with technetium Tc 99m oxidronate after cation levels have normalized (eg, 3 to 5 half-lives of the cation) in patients with high calcium levels caused by coadministration of calcium salts. *Risk D: Consider Therapy Modification*

Technetium Tc 99m Pyrophosphate: Coadministration of Calcium Salts and Technetium Tc 99m Pyrophosphate may alter diagnostic results. Management: Consider performing imaging with technetium Tc 99m pyrophosphate after cation levels have normalized (eg, 3 to 5 half-lives of the cation) in patients with high calcium levels caused by coadministration of calcium salts. *Risk D: Consider Therapy Modification*

Tetracyclines: Calcium Salts may decrease serum concentrations of Tetracyclines. Management: If coadministration of oral calcium with oral tetracyclines cannot be avoided, consider separating administration of each agent by several hours. *Risk D: Consider Therapy Modification*

Thiazide and Thiazide-Like Diuretics: May increase serum concentrations of Calcium Salts. *Risk C: Monitor*

Thyroid Products: Calcium Salts may decrease therapeutic effects of Thyroid Products. Management: Separate the doses of the thyroid product and the oral calcium supplement by at least 4 hours. Monitor for decreased therapeutic effects of thyroid products if an oral calcium supplement is initiated/dose increased. *Risk D: Consider Therapy Modification*

Trientine: Polyvalent Cation Containing Products may decrease serum concentrations of Trientine. Management: Avoid concomitant use of trientine and polyvalent cations. If oral iron supplements are required, separate the administration by 2 hours. For other oral polyvalent cations, give trientine 1 hour before, or 1 to 2 hours after the polyvalent cation. *Risk D: Consider Therapy Modification*

Unithiol: May decrease therapeutic effects of Polyvalent Cation Containing Products. *Risk X: Avoid*

Vitamin D Analogs: Calcium Salts may increase adverse/toxic effects of Vitamin D Analogs. *Risk C: Monitor*

Pregnancy Considerations

Calcium crosses the placenta. The amount of calcium reaching the fetus is determined by maternal physiological changes (IOM 2011).

Calcium requirements are the same in pregnant patients and nonpregnant females (IOM 2011). Information related to use as an antidote in pregnancy is limited. In general, medications used as antidotes should take into consideration the health and prognosis of the mother; antidotes should be administered to pregnant patients if there is a clear indication for use and should not be withheld because of fears of teratogenicity (Bailey 2003).

Breastfeeding Considerations

Calcium is present in breast milk.

The amount of calcium in breast milk is homeostatically regulated and not altered by maternal calcium intake. Calcium requirements are the same in lactating patients and nonlactating females (IOM 2011). According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and the benefits of treatment to the mother.

Dietary Considerations

Dietary reference intake for calcium (IOM 2011):

0 to <6 months: Adequate intake: 200 mg **elemental calcium/**day.

6 to 12 months: Adequate intake: 260 mg **elemental calcium**/day.

1 to 3 years: Recommended dietary allowance (RDA): 700 mg **elemental calcium/**day.

4 to 8 years: RDA: 1,000 mg **elemental calcium/**day.

9 to 18 years: RDA: 1,300 mg **elemental calcium/**day.

19 to 50 years: RDA: 1,000 mg **elemental calcium/**day.

51 to 70 years: RDA:

Females: 1,200 mg **elemental calcium/**day.

Males: 1,000 mg **elemental calcium/**day.

>70 years: RDA: 1,200 mg **elemental calcium/**day.

Pregnancy/Lactating: RDA: Requirements are the same as in nonpregnant or nonlactating females.

Monitoring Parameters

IV: Serum calcium every 4 to 6 hours (during intermittent infusion), every 1 to 4 hours (during continuous infusion), or every 4 hours in patients with renal impairment; albumin, phosphate, and magnesium; vitals and ECG when appropriate. Monitor infusion site.

Calcium channel blocker toxicity, beta-blocker toxicity (off-label use): Monitor hemodynamic response; monitor serum ionized calcium levels every 30 minutes initially, then every 2 hours and maintain ionized calcium ~1.5 to 2 times the ULN; avoid severe hypercalcemia (ionized calcium levels >2 times ULN) (AHA [Lavonas 2023]; Kerns 2007).

Reference Range

Serum total calcium: 8.6 to 10.2 mg/dL (SI: 2.2 to 2.6 mmol/L). **Note:** Due to a poor correlation between the serum ionized calcium (free) and total serum calcium, particularly in states of low albumin or acid/base imbalances, direct measurement of ionized calcium is recommended.

Serum ionized calcium: 4.48 to 4.92 mg/dL (SI: 1.12 to 1.23 mmol/L).

In low albumin states, the corrected **total** serum calcium may be estimated by the following equation (assuming a normal albumin of 4 g/dL [SI: 40 g/L]).

Corrected total calcium (mg/dL) = measured total calcium (mg/mL) + 0.8 [4-measured serum albumin (g/dL)].

or

Corrected total calcium (mmol/L) = measured total calcium (mmol/L) + 0.02 [40-measured serum albumin (g/L)].

Mechanism of Action

Moderates nerve and muscle performance via action potential threshold regulation.

In hydrogen fluoride exposures, calcium gluconate provides a source of calcium ions to complex free fluoride ions and prevent or reduce toxicity; administration also helps to correct fluoride-induced hypocalcemia.

In patients with radium-226 (Ra-226) or strontium-90 internal contamination, calcium gluconate competes for bone binding sites and binds phosphate (REMM 2022).

Pharmacokinetics (Adult Data Unless Noted)

Absorption: Oral: Minimal unless chronic, high doses are given; predominantly in the duodenum and dependent on calcitriol and vitamin D; mean absorption of calcium intake varies with age (infants 60%, prepubertal children 28%, pubertal children 34%, adults 25%); during pregnancy, calcium absorption doubles; calcium is absorbed in soluble, ionized form; solubility of calcium is increased in an acidic environment (IOM 2011); decreased absorption occurs in patients with achlorhydria, renal osteodystrophy, steatorrhea, or uremia

Distribution: Primarily in skeleton (99%)

Protein binding: ~40%, primarily to albumin

Excretion: Primarily feces (75%; as unabsorbed calcium salts); urine (20%) (IOM 2011)

Patient Counseling Points

(For additional information see ”Calcium gluconate: Patient drug information”)

What is this drug used for?

• It is used to treat or prevent low calcium levels.

• It may be given to you for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

• Constipation

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

All products:

• High calcium levels like weakness, confusion, feeling tired, headache, upset stomach and throwing up, constipation, or bone pain

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Injection:

• Flushing, slow or abnormal heartbeat, or signs of low blood pressure like severe dizziness or passing out

• Any redness, burning, pain, swelling, blisters, skin sores, or leaking of fluid where the drug is going into your body

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AR) Argentina: Calcio gluconato | Gluconato calcio | Gluconato de calcio | Solucion de gluconato de calcio al 10% lavimar;
- (AT) Austria: Calcilin;
- (BD) Bangladesh: Calcium | Gultate;
- (BR) Brazil: Gliconato de calcio | Gluconato de calcio;
- (CN) China: Calcium glucon.co | Gai li de;
- (CO) Colombia: Gluconato de calcio;
- (CR) Costa Rica: Gluconato de calcio;
- (DE) Germany: Ca d glukarat | Calciumgluconat fagron | Dobo | Dreisacal;
- (EC) Ecuador: Calcio gluconato | Gluconato de calcio | Gluconato de calcio Ecar;
- (EE) Estonia: Calcii gluconas | Kaltsium glukonaat | Kaltsiumglukonaat;
- (FI) Finland: Calcii gluconas;
- (FR) France: Gluconate de calcium b.braun medical | Gluconate de calcium cooper;
- (HU) Hungary: Calcimusc;
- (IT) Italy: Calcio gluconato | Calcio gluconato monico;
- (JP) Japan: Calcicol;

- (KE) Kenya: Glucal;
- (KR) Korea, Republic of: Daihan calcium gluconate | Jw calcium gluconate;
- (LT) Lithuania: Calcii gluconas | Calcium gluconat | Calcium spofa;
- (LV) Latvia: Calcii gluconas | Calcium gluconase | Kalcija glikonats;
- (MX) Mexico: Bon ker;
- (NZ) New Zealand: Calcium Gluconate B Braun | Radiance Boron;
- (PE) Peru: Calcio gluconato | Glucal | Gluconato de calcio;
- (PH) Philippines: Calcinate | Harson calcium gluconate;
- (PL) Poland: Calcio gluconato | Calcium gluconicum;
- (PT) Portugal: Gluconato de calcio Braun | Gluconato de calcio Labesfal;
- (PY) Paraguay: Gluconato de calcio parafarma | Gluconato de calcio sanderson | Gluconato de calcio veinfar;
- (RU) Russian Federation: Calcium gluconate lekt;
- (TH) Thailand: Calcicon | Calcium;
- (TN) Tunisia: Gluconate de calcium;
- (TW) Taiwan: Calcium gluconated;
- (UA) Ukraine: Calcii gluconas;
- (UY) Uruguay: Gluconato de calcio | Gluconato de calcio icu vita | Gluconato de calcio sanderson;
- (VE) Venezuela, Bolivarian Republic of: Gluconato de calcio;
- (VN) Viet Nam: Growpone;
- (ZA) South Africa: Freeda calcium gluconate | Gulf calcium gluconate | Kalci-300

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